

CV RESEARCH-REPRODUCTION PROJECT · STAGE 2 & STAGE 3

Reproducing Transfer-Learning & Snapshot-Ensemble Diabetic Retinopathy Grading on APTOS 2019

Base paper: Chilukoti et al., BMC Medical Informatics and Decision Making 24:37 (2024)

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github.com/ssshivanshg/dr-grading-reproduction

Where we are in the project

1

Literature survey

10+ papers, gaps identified

2

Pick base paper

Reproducibility checklist

3

Reproduce

Re-run method, explain gap

4

Hypothesis

One change + prediction

5

Implement & ablate

Measure what moved

6

Analyse & write

IEEE paper + defence

▲ TODAY

Stage 1 take-aways (our literature review)

Class imbalance remains unresolved: Severe NPDR & PDR are rare

Accuracy can be misleading; per-class recall and QWK are needed

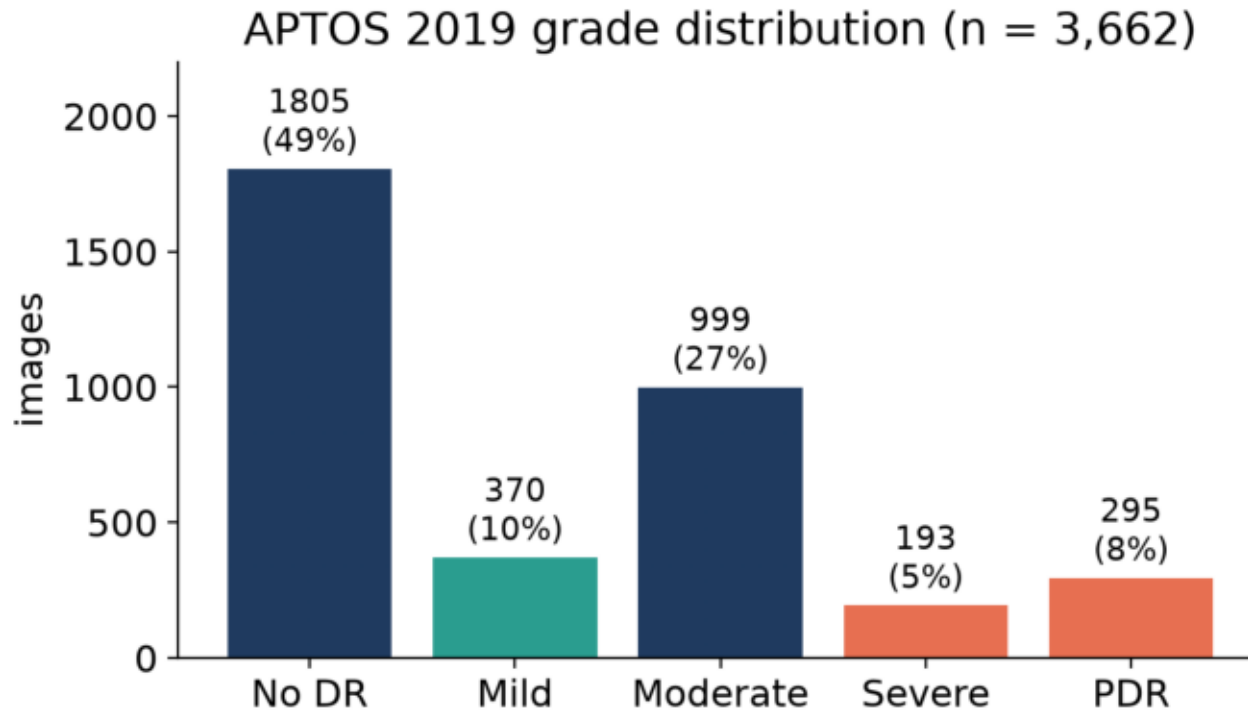
Ordinal nature of grades (0→4) is under-used

Research question we are building towards

Can class-balanced, ordinal-aware learning improve Severe NPDR and PDR recognition while reducing large grading errors?

Stages 2–3 build the reproducible baseline to test it on.

Task: five-class DR grading on APTOS 2019



Dataset

3,662 colour fundus photographs, Aravind Eye Hospital (rural India)

Graded 0–4 by clinicians (ICDR scale)

Split (stratified, seed 42)

70 / 15 / 15 → 2,562 train · 550 val · 550 test

Same proportions the paper used for EyePACS

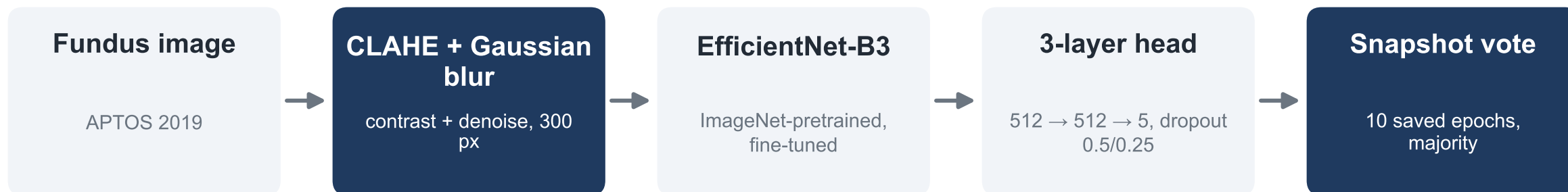
Why it is hard

Severe NPDR = 5%, PDR = 8% of images

Adjacent grades look alike (Mild↔Moderate, Severe↔PDR)

Grades are ordinal, so error size matters → QWK

Base paper: Chilukoti et al. (2024)



Dark boxes = the paper's two claimed contributions

Reported on APTOS 2019	QWK
ImageNet init, single model	0.954
EyePACS init, single model	0.956
EyePACS init + snapshot ensemble	0.967

Our target: 0.954 (ImageNet init). EyePACS pre-training needs 35k extra images, beyond our compute.

Training recipe stated in the paper

Adam, learning rate 1e-3, weight decay 1e-4

Gradient clipping 0.1

Strategy 2: CLAHE + blur at 300×300 , weights saved at 10 intervals

QWK as the primary metric

Why we chose this paper

Same dataset & task

Five-class grading on APTOS 2019, the course-recommended dataset

Ordinal metric

Uses QWK as primary metric, matching our Stage 1 finding

Feasible compute

Transfer learning from one pretrained CNN, no heavy ensemble of models

Leaves our gaps open

No imbalance-aware or ordinal loss, so there's room for one clean change

Strong claim to test

0.954–0.967 QWK is state-of-the-art level, worth verifying

Clear method text

Architecture and optimiser are described well enough to re-implement

Reproducibility checklist

Checklist item	Our answer	Status
Public code?	No repository linked or found; re-implemented from the paper text	✗
Dataset public & size?	APTOS 2019, 3,662 images, ~8.9 GB (full-res mirror on Hugging Face)	✓
Fits free GPU?	Expected yes on Colab T4 at 300 px; locally (M2, 8 GB) only at 224 px	~
Code usable / last commit?	N/A; our code: PyTorch 2.14 + timm 1.0.30, resumable training	—
Metric defined?	QWK (primary) + accuracy, precision, recall, F1, AUC	✓
Split defined?	Only for EyePACS (70/15/15); not for APTOS	~
Hyper-parameters?	Optimiser, lr, decay, clipping given; batch size, loss, epochs, CLAHE params not	~

✓ available ~ partially specified ✗ missing

Filling the gaps: our implementation choices

Not stated in paper	Our choice	Why
Loss function	Cross-entropy	Softmax output implies it
“Gradient clipping 0.1”	Global L2-norm clip at 0.1	Most common reading
CLAHE settings	Clip 2.0, 8×8 tiles, on LAB luminance	Standard; keeps colour
Blur	5 × 5 Gaussian, after CLAHE	Order as written in paper
APTOS split	Stratified 70/15/15, seed 42	Paper's EyePACS proportions
Final or best model?	Report both + both ensembles	Avoid cherry-picking

Red flags in the paper

Table 3: precision = recall = QWK = 0.867 exactly

Constant predictors report macro AUC 0.196 (a constant predictor gives 0.5)

Table 6: accuracy, precision, recall, F1 all within 0.01 on imbalanced data

APTOS split unspecified → possible leakage

We treat 0.954 as a claim to test, not ground truth.

Data audit: APTOS contains duplicate images

123

groups of byte-identical images

30

groups with conflicting grades

29 / 550

test images with a copy in train

± 0.001

QWK change when removed

What we did

SHA-1 hash of every raw image file

Perceptual hashing rejected: every fundus is a bright disc on black, giving thousands of false matches

Re-scored each model on the 521 test images without a training copy

What it means

Leakage does not explain our gap (QWK 0.827 $\rightarrow$ 0.828)

Leaked images were predicted worse (65.5% acc.), because of conflicting labels

Label noise exists in APTOS itself (e.g. the same image graded Moderate and PDR)

Implementation under hardware limits

Setting	Paper	Ours	Reason
Backbone / head / optimiser	EffNet-B3, 3-layer, Adam 1e-3	Identical	—
Pre-processing	CLAHE + blur	Identical	—
Resolution	300 × 300	224 × 224	300 px exceeds 8 GB memory
Epochs / snapshots	60 / 10	30 / 10 (every 3)	~4 min per epoch locally
Batch size	not stated	8	largest without swapping
EyePACS pre-training	yes (for 0.967)	no	35k images, out of budget

Benchmarked on M2

B3 @ 300 px: swaps, 25–45 s / step

B3 @ 224 px, bs 8: 2.8 GB, 12.4 img/s

Mixed precision: no speed-up

Engineering

Pre-processed cache (memory-mapped)

Resumable checkpoints every epoch

Seeded: bit-identical across launches

Colab notebook for exact protocol

Results: reproduction vs. paper

Configuration	Prediction	QWK	Acc.	Macro-F1	AUC	Severe rec.	PDR rec.
Paper settings (lr 1e-3)	single model	0.827	0.771	0.473	0.897	0.00	0.30
Paper settings (lr 1e-3)	10-snapshot vote	0.788	0.767	0.412	0.913	0.00	0.00
Only change: lr 1e-4	single model	0.887	0.815	0.630	0.894	0.31	0.55
Only change: lr 1e-4	10-snapshot vote	0.887	0.822	0.613	0.930	0.10	0.59
Paper (reported)	single / ensemble	0.954 / 0.967	0.942	0.944	—	—	—

-0.127

QWK gap with paper settings

+0.060

QWK from lr change alone

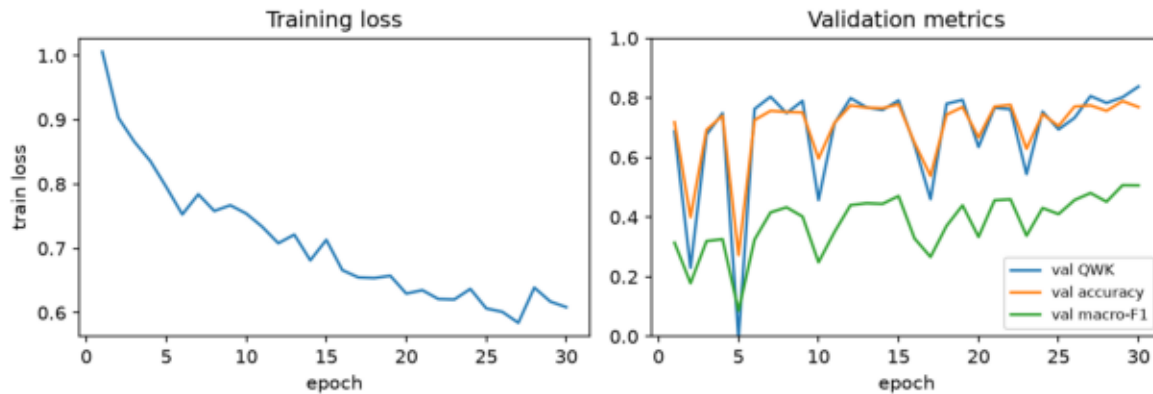
0 → 0.31

Severe NPDR recall

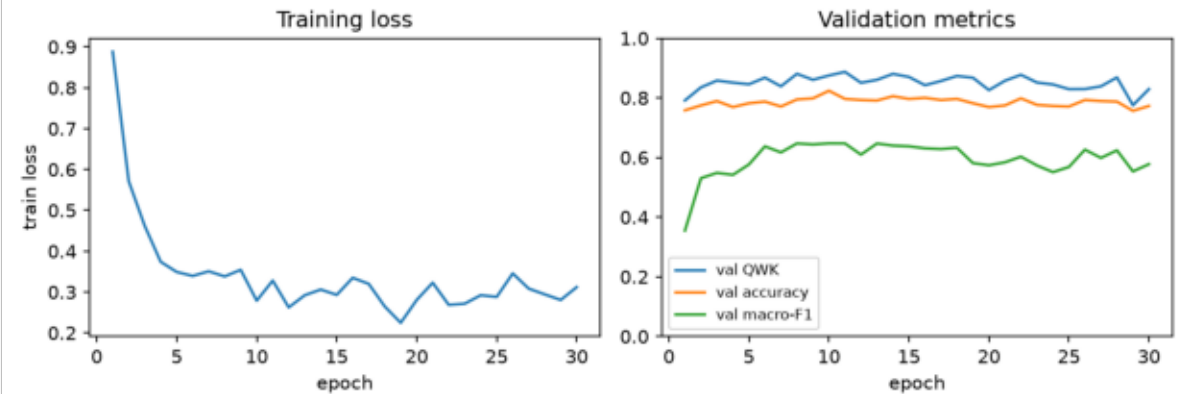
48 → 24

errors of ≥ 2 grades

Why: training at $lr\ 1e-3$ is unstable



Paper settings: $lr\ 1e-3$



Only change: $lr\ 1e-4$

Symptoms

Validation QWK collapses 5 times (down to -0.006 at epoch 5)

Training loss still 0.61 after 30 epochs: under-fitting, not over-fitting

Explanation

Adam at $1e-3$ over-writes pretrained features in full fine-tuning

Small batch (8) adds noisy batch-norm statistics

Paper's own Table 3 shows VGG/ResNet collapsing to class 0

QWK hides minority-grade failure

QWK=0.827

True \ Predicted	No DR	Mild	Moderate	Severe	PDR
No DR	259	6	6	0	0
Mild	7	13	36	0	0
Moderate	3	1	139	0	7
Severe	1	0	15	0	13
PDR	0	4	27	0	13

Paper settings (lr 1e-3)

QWK=0.887

True \ Predicted	No DR	Mild	Moderate	Severe	PDR
No DR	264	7	0	0	0
Mild	7	22	24	2	1
Moderate	1	5	129	12	3
Severe	1	0	11	9	8
PDR	0	4	12	4	24

lr 1e-4

At lr 1e-4: Severe 9 / 29 correct, PDR 24 / 44; large errors halve. Severe remains the hardest grade, which is the target for Stage 4.

Read-out

Severe NPDR: 0 / 29 correct at lr 1e-3

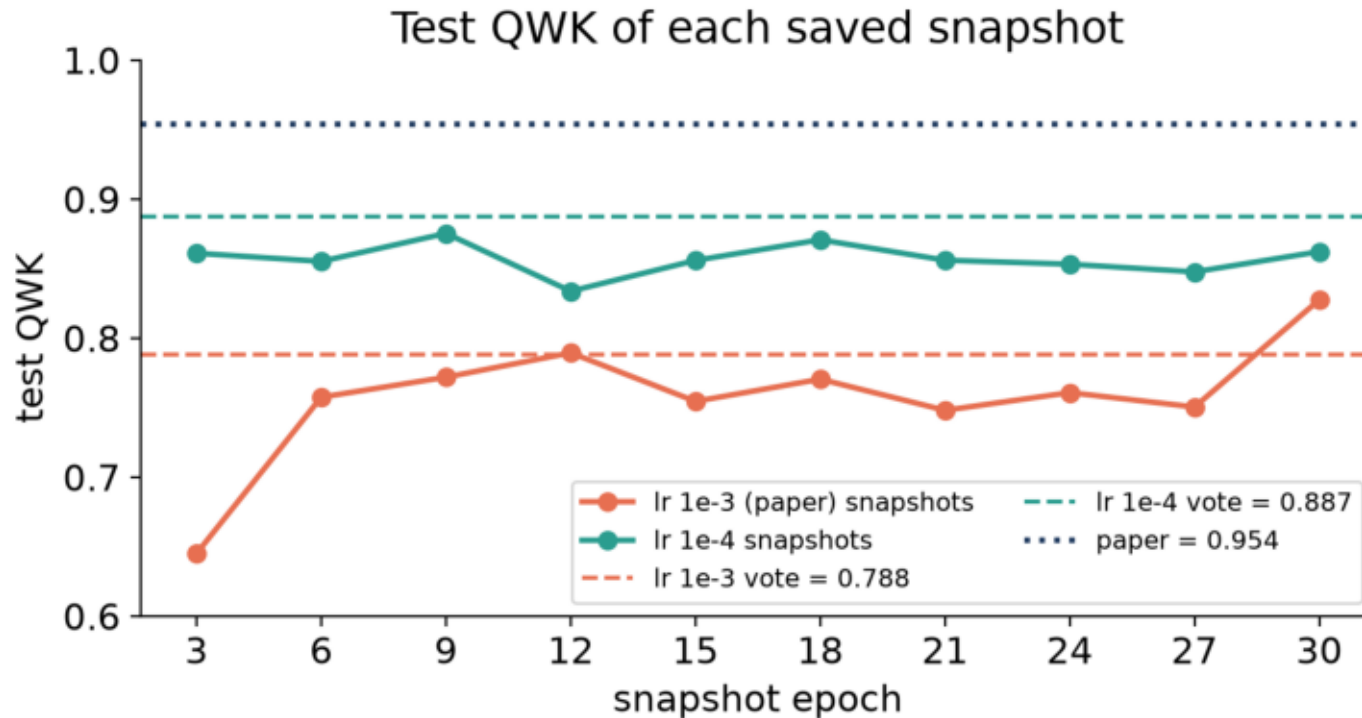
27 / 44 PDR called Moderate

Mild mostly → Moderate

Errors are adjacent, so QWK stays ≈ 0.83

Per-class recall is essential

The snapshot ensemble only works when training is stable



Paper settings (lr 1e-3)

Snapshot QWK 0.65 – 0.79, last one 0.83

Majority vote = 0.788, worse than single model

Severe & PDR recall in vote: 0.00

Stable training (lr 1e-4)

Every snapshot 0.83 – 0.88

Vote = 0.887, above all snapshots

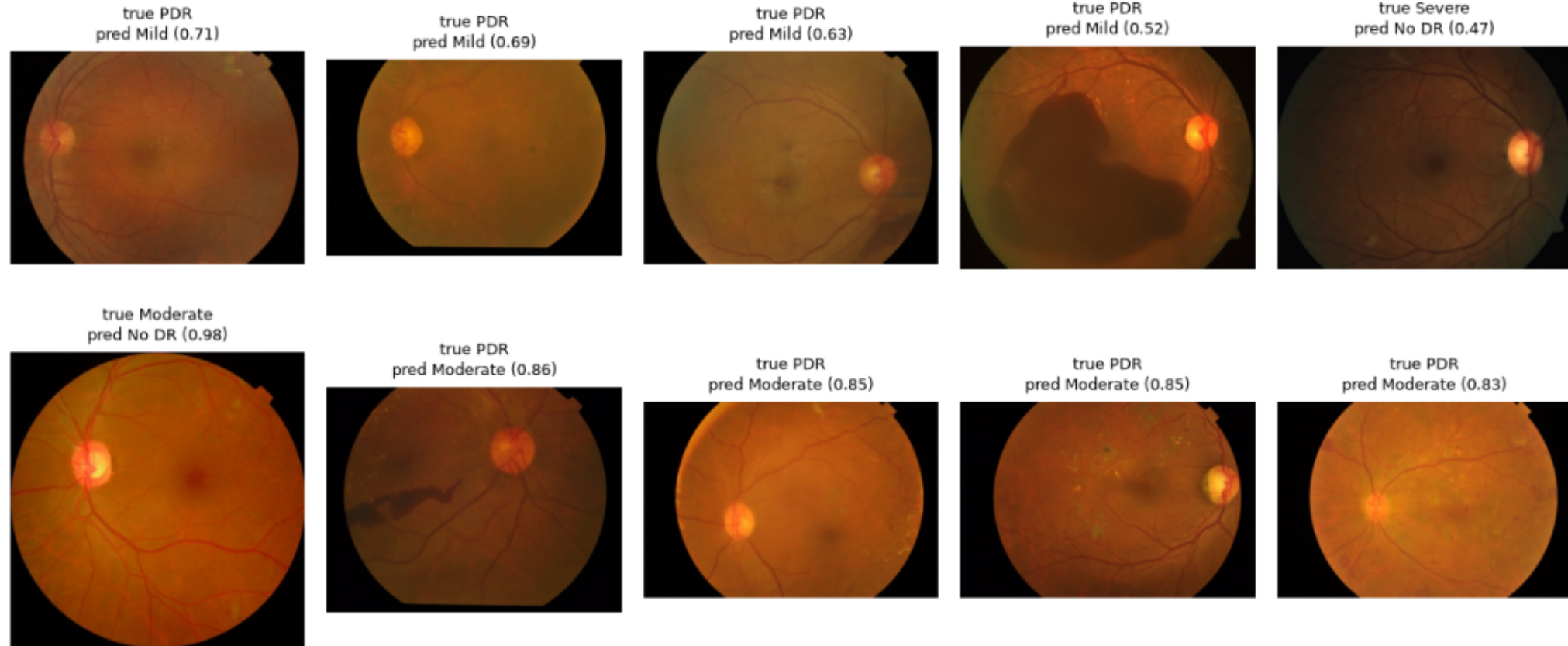
Ensemble helps only with good, diverse members (Huang et al., 2017)

Explaining the reproduction gap (0.827 vs 0.954)

Candidate cause	Evidence	Verdict
Learning rate 1e-3 (paper's setting)	Changing only lr → +0.060 QWK, stable training	Main cause
224 px / 30 epochs (hardware)	Small lesions less visible; not tested yet (Colab notebook ready)	Likely part
Unstated details (split, batch, loss)	Each can move QWK a few points on 550 test images	Possible
Duplicate leakage	QWK 0.827 → 0.828 without the 29 leaked images	Ruled out
Reliability of reported numbers	Internally inconsistent tables (slide 7)	Open question

The single change of learning rate closes about half the gap; the rest is consistent with our lower resolution and shorter schedule.

Failure cases



Mostly PDR → Mild/Moderate (incl. a large pre-retinal haemorrhage); several are dim, low-contrast or cropped. Neovascularisation is hard to see at 224 px.

Limitations (and what we are honest about)

Experimental

Single dataset (APTOS), single seed, no external validation

224 px / 30 epochs instead of 300 px / 60 epochs

EyePACS-pretrained setting (0.967) not reproduced

Only 29 Severe and 44 PDR test images: 1 image $\approx$ 3 recall points

Paper's strategy 1 (150 px) skipped: reported on EyePACS only

Process

Headline run interrupted (disk full) and resumed from epoch 13 checkpoint

The lr 1e-4 run is our diagnostic, not the paper's protocol

No public code, so some choices are our interpretation

No clinical claims: this is a reproduction study

Next: Stage 4 hypothesis (predicted before coding)

Hypothesis

Replacing cross-entropy with a class-balanced, ordinal-aware loss on the stable lr 1e-4 baseline will improve Severe NPDR and PDR recognition and reduce large grading errors.

Mechanism

- Class weights raise the gradient contribution of rare grades
- Ordinal term penalises errors by distance from the true grade
- Targets both gaps from our literature review

Metric	Baseline	Predicted
Severe recall	0.31	+0.10 to +0.20
Minority recall	0.43	increase
Errors ≥ 2 grades	24	< 24
QWK	0.887	0 to +0.02
Accuracy	0.815	may dip 1–2 pts

Summary

1

Reproduced from scratch

No public code. With the paper's settings: QWK 0.827 vs 0.954 claimed.

2

Found the main cause

Lr 1e-3 makes training unstable; Lr 1e-4 alone gives 0.887.

3

Ensemble claim is conditional

Snapshot vote hurts when unstable (0.788) and helps when stable.

4

QWK is not enough

Severe NPDR was never detected at QWK 0.83: report per-class recall.

Code, logs, figures & report: github.com/ssshivanshg/dr-grading-reproduction · Questions?